

**HS-402 ENGINEERING ECONOMICS
G06-INDUSTRIAL-GRADE OMR SCANNER GROUP**



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NAME:

HUZAIFA SHAHBAZ

AQSA SHAH

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**Department of Computer Engineering
HITEC University, Taxila**

Table of Contents

Industrial-Grade OMR Scanner for Efficient Workforce & Process Management	3
ABSTRACT	3
1.INTRODUCTION	3
1.1 INTRODUCTION	3
1.2 OBJECTIVES.....	3
1.3 PROBLEM STATEMENT.....	3
2. PURPOSE OF THE PRODUCT	3
3. MARKET NEED AND INDUSTRIAL APPLICATION	4
4. PRODUCT DESIGN FEATURES	4
4.1 HARDWARE FEATURES	4
4.2 SOFTWARE FEATURES	4
4.3 FORM DESIGN	4
5. TARGET USERS AND BUSINESS SCOPE	4
6. PRODUCT DEVELOPMENT TIMELINE	5
7. COMPETITIVE ANALYSIS.....	5
8. PROJECT ESTIMATION	5
9. CHALLENGES AND RISK MITIGATION	6
10. CONCLUSION.....	6
References	6

Industrial-Grade OMR Scanner for Efficient Workforce & Process Management

ABSTRACT

In the current era of Industry 4.0, in which digital transformation is paramount for productivity and operational efficiency, there still remains a gap on the side of automation for administrative work in manufacturing enterprises. OMR scanners, which are mostly used in the academic sectors, can be effectively redesigned for automating evaluations, surveys, and inspections in large industrial settings. This report looks into the concept, feasibility, and business scope of launching an industrial OMR scanner tailor-made for multinational manufacturing companies. Built to withstand harsh shop-floor conditions and integrated with existing ERP systems, this scanner aims to reduce human error in manual data entry, expedite workforce assessment, and enhance compliance processes. This report provides the technical specification, market demand, use cases, financial projections, and a six-month roadmap for making a successful product launch.

1.INTRODUCTION

1.1 INTRODUCTION

In a fast-paced manufacturing business featuring digitization, data accuracy, and smart automation, even administrative tools can help improve productivity or reduce manual error. Traditionally considered an academic tool, Optical Mark Recognition (OMR) scanners are being redefined for industrial applications. This report proposes the design and product development of an industrial-grade OMR scanner specific to the administrative and operational requirements of multinational manufacturing companies.

1.2 OBJECTIVES

- Analyze the current state of OMR and its germinating technologies for non-academic purposes, particularly in manufacturing administration.
- The study aims to identify operational gaps in employee assessment, training, and compliance documentation that can be addressed using OMR technology.
- Create a prototype of an industrial-grade OMR scanner with advanced features suitable for rugged environments and enterprise integration.
- The proposed solution will be compared to existing generic OMR products to showcase their distinctive industrial value.
- The study evaluates the economic and operational advantages of implementing this product in multinational manufacturing companies, focusing on ROI, efficiency, and scalability.

1.3 PROBLEM STATEMENT

Technology has significantly impacted communication, work, and entertainment, with smartphones and social media platforms enabling constant connectivity. Remote work and virtual meetings have become the norm, offering flexibility and collaboration opportunities. As technology evolves, its influence on society is expected to expand, presenting both challenges and benefits.

2. PURPOSE OF THE PRODUCT

The launch of an industrial-grade OMR scanner aims to:

- Enhance the speed and precision of employee evaluations.
- The goal is to streamline data collection for training, recruitment, feedback, and compliance audits.

- The goal is to replace the error-prone manual data entry systems with a cost-effective and scalable solution.
 - The goal is to improve decision-making by integrating fast analytics efficiently.
- This product addresses the growing demand for data-backed personnel management systems in diverse industrial environments worldwide.

3. MARKET NEED AND INDUSTRIAL APPLICATION

Manufacturing companies prioritize robotics, sensors, ERP systems, and IoT, but administrative tools often lag behind in innovation, with OMR scanning potentially supporting this gap.

- **RECRUITMENT AND SKILL TESTING:**
Factories frequently use written assessments for hiring operators, engineers, and technicians, utilizing OMR to minimize time and human error in evaluation.
- **EMPLOYEE TRAINING AND CERTIFICATIONS:**
Mandatory safety and quality training often incorporates MCQ-based evaluations, which can be automated using an OMR scanner for score processing across thousands of employees.
- **QUALITY AND COMPLIANCE AUDITS:**
Inspection teams frequently create checklists or report that can be standardized using OMR sheets.
- **CUSTOMER AND VENDOR FEEDBACK:**
An OMR scanner is a valuable tool for companies that still rely on printed forms due to language barriers or local connectivity issues.
- **SURVEYS AND INTERNAL POLLING:**
The study aims to facilitate the swift collection of data from labor-intensive departments without the use of smartphones or computers.

4. PRODUCT DESIGN FEATURES

4.1 HARDWARE FEATURES

- HIGH SPEED SCANNING(UP TO 300 SHEETS/MIN)
- DUAL-SIDE SCANNING WITH AUTOMATIC PAGE FEED
- DUSTPROOF AND FACTORY-FLOOR COMPLIANT CASING(IP54 PROTECTION)
- USB 3.0AND ETHERNET CONNECTIVITY
- LED STATUS INDICATOR
- EXPANDABLE PAPER TRAY FOR UP TO 500 SHEETS

4.2 SOFTWARE FEATURES

- SMART ALIGNMENT DETECTION AND AUTO CORRECTION
- OCR AND BARCODE READER SUPPORT
- BATCH PROCESSING MODE
- EPR INTEGRATION APIs (SAP,ORACLE)
- DATA EXPORT IN PDF,XML.
- MULTILINGUAL INTERFACE SUPPORT DASHBOARD FOR ANALYTICS AND TREND TRACKING

4.3 FORM DESIGN

COMPATIBALE WITH CUSTOMIZED FORMS ARE:

- MULTIPLE CHOICE TEST
- FEEDBACK GRIDS
- YES/NO SURVEYS
- SCORED AUDIT CHECKLISTS

5. TARGET USERS AND BUSINESS SCOPE

END USERS:

The HR departments are responsible for recruitment and training, safety and compliance officers, floor managers and line supervisors, quality auditors, and vendor management teams.

GEOGRAPHICAL SCOPE:

The company operates manufacturing plants and offices in various regions, including South Asia, Southeast Asia, Europe, and North America, focusing on industries like automotive, textiles, electronics, pharmaceuticals, and consumer goods.

BUSINESS BENEFITS:

The study demonstrates a significant reduction in assessment **processing time by 80%**, **improved accuracy by 95%**, and a **potential ROI within 12-18 months** due to operational efficiency.

6. PRODUCT DEVELOPMENT TIMELINE

Table 6.1:TABLE OF PRODUCT DEVELOPMENT

MONTH	MILESTONE
1	MARKET ANALYSIS AND REQUIREMENT SURVEY
2	HARDWARE PROTOTYPING AND SOFTWARE ARCHITECTURE
3	INITIAL TESTING AND DEBUGGING
4	BETA VERSION RELEASE AND USER TESTING
5	FINAL PRODUCT DESIGN, CERTIFICATIONS
6	MARKETING AND DOCUMENTATION AND TESTING

7. COMPETITIVE ANALYSIS

Table 7.2:TABLE OF COMPETITIVE ANALYSIS

FEATURE	OUR PRODUCT	GENERIC OMR
DUSTPROOF HOUSING	YES	NO
ERP/API INTEGRATION	YES	LIMITED
DUAL-SIDE SCAN	YES	SOMETIMES
FACTORY FLOOR USE	DESIGNED FOR IT	NO
MULTILINGUAL SOFTWARE	YES	RARE

8. PROJECT ESTIMATION

Table 8.3:TABLE OF PROJECT ESTIMATION

ITEM	COST	INTEREST RATE	TOTAL COST
R&D PROTOTYPING	14,000,000 PKR	2,800,000 PKR	16,800,000 PKR
MANUFACTURING SETUP	28,000,000 PKR	5,600,000 PKR	33,600,000 PKR
MARKETING AND DISTRIBUTION	8,400,000 PKR	1,680,000 PKR	10,080,000 PKR
TOTAL	50,400,000 PKR	10,080,000 PKR	60,480,000 PKR

THE FIGURE IS SHOWN BELOW OF PROJECT ESTIMATION

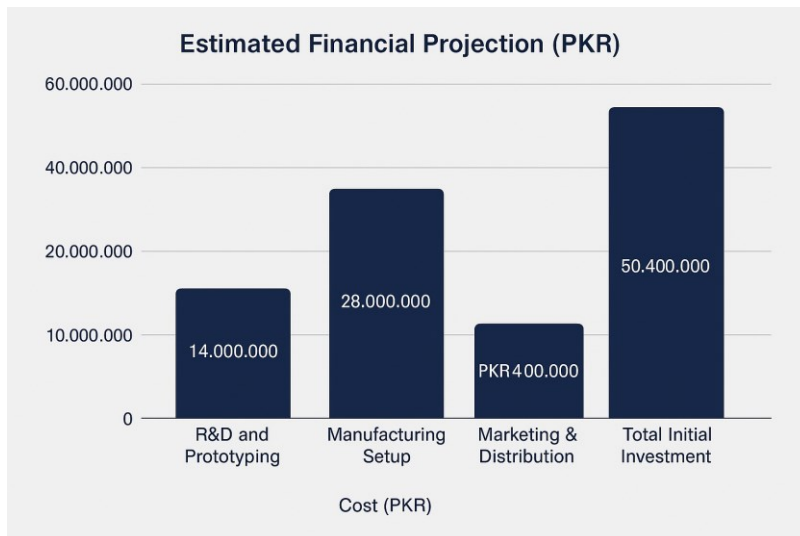


Figure 8.1: ESTIMATED FINANCIAL PROJECTION AND COST(PKR) GRAPH

9. CHALLENGES AND RISK MITIGATION

Table 9.4: TABLE OF CHALLENGES AND RISK MITIGATION

CHALLENGES	RISK MITIGATION
ADOPTION RESISTANCE	PILOT ROLLOUT WITH HR AND QUALITY DEPARTMENTS
LANGUAGE/LOCALIZATION ISSUES	MULTILINGUAL TEMPLATES AND UI
COMPATIBILITY WITH EXISTING SYSTEMS	DEVELOP OPEN API AND MODULAR SDK
ENVIRONMENT DURABILITY	IP-RATED HOUSING AND INDUSTRIAL TESTING

10. CONCLUSION

In conclusion, the industrial-grade OMR scanner offers a cost-effective, efficient solution for administrative and compliance functions in large-scale manufacturing. It integrates minimally with digital systems, addressing operational gaps and future scalability. Launched within a 6-month timeframe, it's strategically advantageous in the hybrid industrial ecosystem.

References

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